

ICT703 Programming

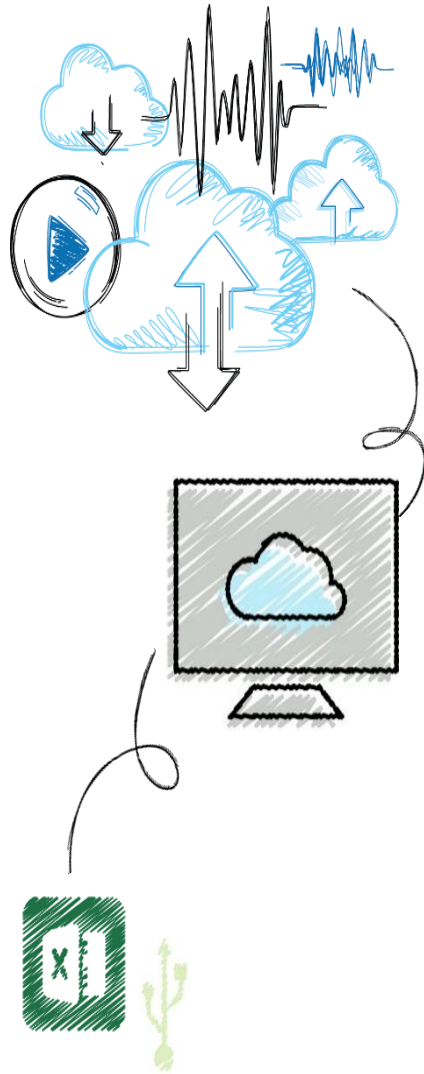
Module 4 Topic 2

Lists



Objectives

1. Create a list
2. Retrieve an item from a list
3. Change or remove items from a list





Programming Concepts

Four control flow

- Sequence
- Iteration
- Selection
- Abstraction – methods/classes

Data



The Structure of Lists

- A list is a very useful data type as it can be used to store lots of information – especially good when you do not know how much upfront.
- Lists can have duplicates
- An empty list is []
- A string is a special form of a list (list of characters)

```
studentList = []
```

```
studentList = ["Fred", "Joan", "Peter", "Fred", "Xiao Ming"]
```

```
len(studentList)
```

```
5
```



Add or change a list

- The easiest way to add to a list is to append values to a list at the end

```
studentList = ["Fred", "Joan", "Peter", "Xiao Ming"]
```

```
studentList.append("Mary")
```

```
print(studentList)
```

```
['Fred', 'Joan', 'Peter', 'Xiao Ming', 'Mary']
```

- The easiest way to delete an item is to remove values

```
studentList.remove("Fred")
```

```
print(studentList)
```

```
['Joan', 'Peter', 'Xiao Ming', 'Mary']
```



Access elements in a list

- We can use the subscript operator (like strings) to access elements of the list

```
studentList = ["Fred", "Joan", "Peter", "Xiao Ming"]
```

```
print(studentList[0])
```

Fred

- We can get a range of items

```
print(studentList[1:3])
```

['Joan', 'Peter']



List Methods

Method	Description
<code>append()</code>	Adds an element at the end of the list
<code>clear()</code>	Removes all the elements from the list
<code>copy()</code>	Returns a copy of the list
<code>count()</code>	Returns the number of elements with the specified value
<code>extend()</code>	Add the elements of a list (or any iterable), to the end of the current list
<code>index()</code>	Returns the index of the first element with the specified value
<code>insert()</code>	Adds an element at the specified position
<code>pop()</code>	Removes the element at the specified position
<code>remove()</code>	Removes the item with the specified value
<code>reverse()</code>	Reverses the order of the list
<code>sort()</code>	Sorts the list



Example – Sorting M&Ms

```
mms = ['red', 'blue', 'green', 'blue', 'green']           #My M&Ms
redBowl = []                                             #Initialise the bowls
blueBowl = []
greenBowl = []
leftoverBowl = []

for eachMM in mms:                                     #Loop through each M&M and check the colour. Put into the appropriate bowl
    if eachMM == 'red':
        redBowl.append(eachMM)
    elif eachMM == 'blue':
        blueBowl.append(eachMM)
    elif eachMM == 'green':
        greenBowl.append(eachMM)
    else:
        leftoverBowl.append(eachMM)

print('Red Bowl', redBowl)
print('Blue Bowl', blueBowl)
print('Green Bowl', greenBowl)
print('Left Over Bowl', leftoverBowl)
```